

Mapping Law & Plank (2023) onto mizer

A mizer reimplementation of Law & Plank (2023), *Fish and Fisheries*

Law, R. & Plank, M.J. (2023) *Fishing for biodiversity by balanced harvesting*. **Fish and Fisheries** 24, 1–17. doi:[10.1111/faf.12705](https://doi.org/10.1111/faf.12705). Equation numbers below are those of the paper (preprint doi:[10.1101/2021.06.27.450047](https://doi.org/10.1101/2021.06.27.450047), v3).

Law & Plank write the dynamics in log body mass $x = \ln(w/w_0)$ with $w_0 = 1$ g and state density $u_i(x, t)$; mizer works in w with number density $N_i(w) = \phi_i(w)$, and $u = w \phi$. Dividing Eq. (A.1) through by w turns it term by term into mizer's equation, so the two are the same PDE.

Units throughout: grams, years, square metres of sea surface. There is no κ -style rescaling — densities are per m^2 exactly as in the paper.

Term by term

Paper	mizer
Eq. (A.1a) somatic growth $-d/dx[\epsilon_i g_i u_i]$	<code>e_growth</code> , with <code>psi = 1 - eps_i</code>
Eq. (A.2) mass-specific intake g_i	<code>alpha * encounter</code> ; <code>search_vol = gamma w^q</code>
— search-rate coefficient A_i , exponent α	<code>gamma = A_i / (6 sigma)</code> , <code>q = 0.85</code>
— conversion efficiency $\kappa = 0.1$	<code>alpha = 0.1</code> (mizer's assimilation efficiency)
— no maximum intake rate	<code>FeedingLevel</code> overridden to return 0
— no metabolic cost	<code>ks = k = 0</code> , so <code>metab = 0</code>
Eq. (A.3) predation mortality d_i	<code>pred_mort</code> (standard mizer)
Eq. (A.4) box feeding kernel, height $1/(6 \sigma)$	<code>pred_kernel_type = "box"</code> , <code>ppmr_min = e^(beta-3 sigma) = 10</code> , <code>ppmr_max = e^(beta+3 sigma) = 1e5</code> ; the height is folded into <code>gamma</code> because mizer's box kernel has height 1
Eq. (A.6) larval mortality μ_l	<code>ext_mort</code> (static in time)
Eq. (A.7) background mortality μ_b	added by the custom <code>Mort</code> function (food-dependent, see below)
Eq. (A.8) allocation $1 - \epsilon_i$	<code>psi = maturity * repro_prop</code> ; <code>w_mat25 = w_mat/3^(1/rho_m)</code> gives mizer's maturity exponent <code>U = rho_m = 15</code> ; <code>m - n = rho_inf = 0.2</code> gives <code>repro_prop = (w/w_max)^0.2</code>
Eq. (A.9) egg production R_i	<code>mizerRDI</code> with <code>erepro = 2</code> <code>eps_R = 0.4</code> (mizer carries a factor 1/2 for the sex ratio that the paper does not)
— no stock-recruitment relationship	<code>RDD = "noRDD"</code>

Paper	mizer
Eq. (A.1f) diffusion $(1/2) \frac{d}{dx}(e^{-x} \frac{d}{dx}[\epsilon_i G_i u_i])$	<code>use_predation_diffusion = TRUE</code> ; mizer's <code>D</code> omits the <code>eps_i</code> factor, so the custom <code>Diffusion</code> function multiplies it back in
Eq. (A.11) plankton logistic + immigration	custom <code>resource_dynamics = "lpResource"</code>
Eqs (A.12)–(A.14) $r(x)$, $a(x)$, $I(x)$	<code>resource_rate</code> , <code>resource_capacity</code> , and <code>other_params\$lp_immigration</code> , each divided by <code>w</code> to convert the paper's log-density to mizer's per-mass density
Appendix C θ	interaction matrix (0.5 on the diagonal, 0.2 off) and <code>interaction_resource = 1</code>
Eq. (2.3) B_i	<code>lp_biomass()</code>
Eq. (2.4) P_i incl. boundary terms	<code>lp_production()</code>
Eqs (2.7)–(2.9) the three rules	custom <code>FMort = "lpFMort"</code>

Checks that the mapping is exact

`tests/test_model.R` checks, against hand-coded integrations of the paper's equations on the model's own grid:

- the encounter rate against Eq. (A.2);
- the predation mortality against Eq. (A.3);
- `psi` against Eq. (A.8);
- the plankton carrying capacity against Eq. (A.13) and the resulting total primary production ($\sim 4.4e3 \text{ g m}^{-2} \text{ yr}^{-1}$ against the paper's "region of 4000");
- the resource fixed point of `lpResource` against the root of Eq. (A.11);
- `lp_production()` against a finite-difference evaluation of Eq. (2.4).

The diffusion term

Eq. (A.1f) is written in log mass as

$$\frac{1}{2} \frac{\partial}{\partial x} \left(e^{-x} \frac{\partial}{\partial x} [\epsilon_i G_i u_i] \right),$$

with G_i (Eq. A.10) the second moment of prey mass. mizer works in w and adds $\frac{1}{2} \partial^2 (D_i N_i) / \partial w^2$ with D_i in $\text{g}^2 \text{ yr}^{-1}$. The two are the same term. Writing $B = \epsilon_i G_i u_i$ and using $\partial/\partial x = w \partial/\partial w$ and $e^{-x} = 1/w$ (with $w_0 = 1 \text{ g}$),

$$\frac{1}{2} \frac{\partial}{\partial x} \left(\frac{1}{w} \frac{\partial B}{\partial x} \right) = \frac{1}{2} w \frac{\partial}{\partial w} \left(\frac{1}{w} \cdot w \frac{\partial B}{\partial w} \right) = \frac{1}{2} w \frac{\partial^2 B}{\partial w^2},$$

and the whole of Eq. (A.1) is an equation for $u = w\phi$, so dividing through by w leaves $\frac{1}{2} \partial^2 B / \partial w^2$. Since $G_i = D_i/w$ and $u_i = w\phi_i$, we have $B = \epsilon_i D_i \phi_i$, which is mizer's term with the ϵ_i factor that `mizerDiffusion()` omits and `lpDiffusion()` restores. The same division by w

turns Eq. (A.1a) into mizer's advection term, so the correspondence is term-by-term, not just in aggregate.

Ambiguities in the paper, and how they are resolved here

1. **Eq. (A.3) as printed evaluates the predator's search area at the prey's body mass** — $A_j (w_0 e^x)^{\alpha_j}$, where x indexes the prey. Eq. (A.2) uses the consumer's own size, so this is a typo; mizer uses the predator's size and this implementation follows mizer.
2. **Eq. (A.7)** — prose and formula disagree; see `mu_b.md`. This one changes the results qualitatively, and is exposed as the `mu_b_form` option.
3. **Table 1 contradicts the text on x_a and x_I .** The table glosses $w_0 e^{-x_a}$ as 0.001 g, which with $a(x) = a_0 (w_0 e^{x-x_a})^{1-\lambda}$ would put the carrying capacity at $2 \times 10^9 \text{ m}^{-2}$ at 1 mg rather than the stated 2000 m^{-2} . The text's $x_a = \log(0.001)$ gives $a(w) = a_0 (w/0.001)^{1-\lambda}$, which does put it at 2000 m^{-2} at 1 mg and integrates to a total primary production of $4.4 \times 10^3 \text{ g m}^{-2} \text{ yr}^{-1}$ against the paper's "region of 4000". The text is therefore taken to be correct.
4. **The invading spectrum.** Appendix B says only "started at a low density on a power law, the egg density being 0.002 m^{-2} ". Taken here as the paper's own state variable, i.e. the log-density $u(x_0) = 0.002 \text{ m}^{-2}$, so $\phi(w_{\text{egg}}) = 2 \text{ m}^{-2} \text{ g}^{-1}$, on a power law $u \sim w^{-(\lambda-1)}$ matching the plankton spectrum. Only the invasion *outcome* depends on this, and only weakly: an invader that establishes does so over tens of generations.

Where this deviates from the paper, and why

1. **The plankton time step.** The paper integrated Eq. (A.11) with explicit Euler at `dt = 0.002 yr`. `lpResource()` instead uses the closed-form solution of the same Riccati equation over a step, which has the same fixed points but is unconditionally stable, so the time step is not tied to the fastest-dividing cells (r reaches $\sim 320 \text{ yr}^{-1}$ at $1 \text{e-}10 \text{ g}$). `tests/test_convergence.R` checks the two agree to 0.02% at `dt = 0.002`, and that `dt = 0.01` and `dt = 0.002` agree to 0.3%. Assembly is run at `dt = 0.01`; the finished ecosystems are then relaxed and all fishing runs are done at the paper's `dt = 0.002`.
2. **The particular ecosystems differ.** The paper does not report the random seed or the realised w_{max} and μ_{egg} of its 15 species, so the assembly procedure of Appendix B is reproduced but not the specific assemblage. The figures are therefore qualitatively, not numerically, comparable.
3. **Fig. 2a is empirical data** (an Ecopath model of the West Scotland shelf, Alexander et al. 2015) and is not reproduced.

Page https://gustavdelius.github.io/balanced-harvest/mapping.html of the site accompanying the mizer reimplement of Law & Plank (2023). Generated by `make_pdfs.py`.